OUTPRO.INDIA

Corporate Digital Presence Platform

COMPLETE TECHNICAL DOCUMENTATION

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Prepared by	Outpro.India Technical Team
Reviewed by	Senior Architect & DevOps Lead

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1. System Architecture

Outpro.India is a Jamstack-style platform combining static site generation, serverless API routes, and a containerised REST API to deliver a corporate digital presence system with an integrated admin panel.

1.1 High-Level Architecture

The platform is composed of three independent, deployable tiers that communicate through well-defined HTTP APIs and webhooks.

Tier	Technology	Responsibility
Frontend	Next.js 14 · Vercel	Public website, admin panel, SSG/ISR pages
Backend	Node.js · Express · AWS ECS Fargate	REST API, business logic, auth, CRM integration
Data	Supabase (PostgreSQL)	Leads, sessions, audit log, media metadata
Content	Sanity.io (Headless CMS)	Portfolio, services, team, testimonials
CDN / WAF	Cloudflare Pro	Global delivery, WAF, DDoS protection, rate limits
Storage	Supabase Storage	Media assets with ClamAV virus scanning pipeline
Email	Resend API	Transactional email: lead confirmation, admin alerts
CRM	HubSpot / Zoho	Lead sync with exponential-backoff retry

1.2 Request Flow

Every public request follows this path from browser to data store:

1. Browser sends HTTPS request to outpro.india or api.outpro.india

2. Cloudflare WAF evaluates the request: checks against OWASP ruleset, rate limits, and bot signatures. Blocked requests are rejected here — they never reach the origin.

3. Cache HIT: Cloudflare serves the cached SSG page (static HTML/JS/CSS) directly from the nearest PoP. No origin request is made.

4. Cache MISS: Cloudflare forwards the request to Vercel (frontend) or the AWS Application Load Balancer (backend API).

5. Vercel serves Next.js pages: SSG pages are served from Vercel's edge cache. API routes (contact, revalidate) execute as Vercel serverless functions.

6. AWS ALB receives API requests and distributes them across 2–6 ECS Fargate tasks running the Node.js/Express application.

7. The Express application validates the JWT, checks role permissions, validates the request body (Zod), and executes the business logic.

8. The application reads or writes to Supabase PostgreSQL, fetches content from Sanity CDN, or calls external services (Resend, HubSpot).

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9. The response is serialised into the standard JSON envelope and returned through the ALB → Cloudflare → browser chain.

1.3 Security Layers

Layer	Control	Technology
0 – Network	WAF, DDoS, bot management	Cloudflare Pro
1 – Transport	TLS 1.3 end-to-end, HSTS preload	ACM + ALB + Cloudflare
2 – Application	CSP headers, XSS sanitisation (DOMPurify), CSRF	Helmet.js + Next.js headers
3 – Authentication	JWT HS256 + HttpOnly cookie + per-JTI revocation	jsonwebtoken + Secrets Manager
4 – MFA	TOTP (Google Authenticator compatible)	speakeasy
5 – Authorisation	Role-based (super_admin / editor / viewer)	Custom middleware
6 – API	Rate limiting, reCAPTCHA v3, Zod validation	Upstash Redis + Google
7 – Data	Row-Level Security, AES-256 at rest	Supabase + KMS
8 – Supply chain	Dependency audit, container scanning	pnpm audit + Trivy + CodeQL

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2. Tech Stack

2.1 Frontend

Package	Version	Purpose
Next.js	14.2.x	App Router, SSG, ISR, API routes, image optimisation
React	18.3.x	Component model, Suspense, server components
TypeScript	5.4.x	Type safety across the entire codebase
Tailwind CSS	3.4.x	Utility-first styling with custom design tokens
Framer Motion	11.x	Animations: hero, counters, page transitions, carousels
TanStack Query	5.x	Server state management, caching, optimistic updates
Zustand	4.x	Client state: auth, toasts, UI flags
React Hook Form	7.x	Form state, validation integration
Zod	3.x	Runtime schema validation (mirrors backend rules)
Embla Carousel	8.x	Touch-friendly testimonial carousel
DnD Kit	6.x / 8.x	Drag-and-drop testimonial reorder in admin
Lucide React	0.395.x	Icon library
jose	5.x	JWT verification in Next.js Edge middleware

2.2 Backend

Package	Version	Purpose
Node.js	20 LTS	Runtime
Express	4.18.x	HTTP framework, middleware pipeline
pg (node-postgres)	8.11.x	PostgreSQL client with connection pooling
bcrypt	5.1.x	Password hashing (cost factor 12)
jsonwebtoken	9.x	JWT signing and verification
speakeasy	2.x	TOTP MFA secret generation and verification
Zod	3.x	Server-side input validation (all 14 schemas)
Helmet	7.x	HTTP security headers
Winston	3.x	Structured JSON logging with daily rotation
Multer	1.4.x (LTS)	Multipart file upload handling
file-type	19.x	Magic-byte MIME validation (not extension)
DOMPurify + jsdom	3.x / 24.x	HTML sanitisation to prevent XSS
Resend	3.x	Transactional email delivery
Axios	1.7.x	HTTP client for external API calls (HubSpot, reCAPTCHA)

2.3 Infrastructure

Service	Tier	Monthly Cost (USD)

Vercel	Pro (1 seat)	~$20
AWS ECS Fargate	2 tasks × 0.5 vCPU × 1 GB	~$18
AWS ALB	1× ap-south-1	~$18
AWS NAT Gateways	3× AZ (HA)	~$100
Supabase	Pro tier	$25
Cloudflare	Pro	$20
AWS Secrets Manager	1 secret	~$0.40
AWS CloudWatch	Logs + metrics	~$5
AWS S3 (backups + replication)	~5 GB	~$0.75
Total estimate		~$207 / month

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3. Installation Guide

3.1 Prerequisites

Requirement	Minimum Version	Notes
Node.js	20.0.0	Use nvm to manage versions: nvm use 20
pnpm	9.0.0	Install: npm install -g pnpm@9
Docker	24.0	Required for local PostgreSQL and Redis
Docker Compose	2.20	Bundled with Docker Desktop
Git	2.40	
AWS CLI	2.15	For deployment only; optional for local dev
Terraform	1.7.0	For infrastructure provisioning only

3.2 Local Development Setup

Step 1: Clone the repository

```
git clone https://github.com/outpro-india/outpro-platform.git
cd outpro-platform
```

Step 2: Start database services

```
docker compose up postgres redis -d
```

This starts PostgreSQL 16 on port 5432 and Redis 7 on port 6379 using the configuration in docker-compose.yml. Wait 10–15 seconds for both to report healthy.

Step 3: Configure environment variables

```
cp backend/.env.example backend/.env.local
cp frontend/.env.example frontend/.env.local
```

Edit both .env.local files. At minimum set these values for local development:

Variable	Where	How to generate
DATABASE_URL	backend	postgresql://outpro:outpro_dev_password@localhost:5432/outpro
JWT_SECRET	backend	openssl rand -hex 64
REVALIDATION_SECRET	backend + frontend	openssl rand -hex 32
RECAPTCHA_SECRET_KEY	backend	Google reCAPTCHA Admin Console
NEXT_PUBLIC_RECAPTCHA_SITE_KEY	frontend	Google reCAPTCHA Admin Console
RESEND_API_KEY	backend	resend.com → API Keys
SUPABASE_URL	backend + frontend	Supabase Project Settings → API
SUPABASE_SERVICE_ROLE_KEY	backend	Supabase Project Settings → API

Step 4: Run database migrations

```
cd backend && pnpm install --frozen-lockfile
pnpm migrate
```

The migration runner applies all SQL files in migrations/ in alphabetical order and tracks applied versions in schema_migrations. On a fresh database, this takes ~5 seconds.

Step 5: Seed development data (optional)

```
psql $DATABASE_URL -f backend/migrations/003_seed_development.sql
```

> ⚠ **Warning**
> The seed migration is guarded — it will throw an error if run against a production database. Default credentials: arjun@outpro.india / Outpro@Dev2026!

Step 6: Start the backend API

```
pnpm dev   # Runs on http://localhost:5000
```

The API server starts with tsx watch for hot reload. Verify with:

```
curl http://localhost:5000/api/v1/health
```

Step 7: Start the frontend

```
cd ../frontend && pnpm install --frozen-lockfile
pnpm dev   # Runs on http://localhost:3000
```

Next.js starts in development mode with fast refresh. Open http://localhost:3000 in your browser.

3.3 Production Deployment

One-time infrastructure setup

1. Bootstrap Terraform state: bash devops/scripts/bootstrap-tfstate.sh
2. Configure all GitHub Actions secrets (see README.md → First-Time Setup).
3. Provision infrastructure: cd devops/terraform && terraform init && terraform apply
4. Set real secret values in AWS Secrets Manager (see README.md → Step 5).
5. Run database migrations against the Supabase production database.

Subsequent deployments

All subsequent deployments are fully automated via GitHub Actions:

- Push to main branch → backend-deploy.yml runs and deploys to ECS production.
- Push to main branch → frontend-deploy.yml runs and deploys to Vercel production.
- Terraform changes in devops/terraform/ → terraform.yml runs plan on PR, apply on merge.

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4. User Manual

This chapter covers the public-facing website. For admin panel instructions, see Chapter 5.

4.1 Navigating the Website

The Outpro.India website consists of six public pages accessible from the top navigation bar on any device.

Page	URL	Content
Home	/	Hero, statistics, services preview, portfolio highlights, testimonials
About	/about	Company story, team, values, awards and recognition
Services	/services	Overview of all six service areas with deliverables and pricing
Portfolio	/portfolio	Filterable project gallery with category filter and pagination
Testimonials	/testimonials	Client reviews with video testimonials and rating breakdown
Contact	/contact	Contact form, office information, Google Maps location

4.2 Submitting a Contact Form

6. Navigate to https://www.outpro.india/contact
7. Enter your Full Name (required, 2–120 characters).
8. Enter your Business Email (required, must be a valid email address).
9. Optionally enter Company Name, Phone Number (E.164 format: +919876543210), Service Interest, and Budget Range.
10. Enter your Message (required, 10–2000 characters). Describe your project, goals, and timeline.
11. Click Send Message. The form is protected by Google reCAPTCHA v3 (automatic, no checkbox required).
12. On success, a confirmation message will appear. You will also receive a confirmation email to the address provided.

> **ℹ Note**
> Submissions are rate-limited to 100 per minute per IP address. If you receive a rate limit error, wait 60 seconds and try again.

4.3 Browsing the Portfolio

The Portfolio page shows all published projects. You can filter by category using the pill buttons above the grid.

- Click a category pill (All, Web Dev, Branding, Mobile, UI/UX, Marketing, SEO) to filter the grid.
- Click any project card to open the case study page with full project details, results, and gallery.
- Use the pagination controls at the bottom to navigate through multiple pages (9 projects per page).

4.4 Viewing Testimonials

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The Testimonials page displays all client reviews with filter and sort options.

- Use the filter bar to show: All reviews, Featured only, Video testimonials only, or 5-star reviews only.
- For video testimonials, click the play button on the thumbnail to open the video in a modal player.
- The star rating breakdown chart at the top shows the distribution of all ratings.

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5. Admin Guide

5.1 Roles and Permissions

The admin panel has three roles. Every user has exactly one role.

Capability	super_admin	editor	viewer
View dashboard KPIs	✓	✓	✓
View leads	✓	✓	✓ (read only)
Update lead status	✓	✓	✗
Delete leads	✓	✗	✗
Create / update portfolio projects	✓	✓	✗
Delete portfolio projects	✓	✗	✗
Create / update testimonials	✓	✓	✗
Delete testimonials	✓	✗	✗
Reorder testimonials (drag-drop)	✓	✓	✗
Upload media assets	✓	✓	✗
Delete media assets	✓	✗	✗
Manage admin users	✓	✗	✗
View audit log (full IP)	✓	✗	✗
View audit log (redacted IP)	✓	✓	✓

5.2 Logging In

13. Navigate to https://www.outpro.india/login
14. Enter your admin email address and password.
15. If your account has MFA enabled: enter the 6-digit code from your authenticator app.
16. Click Sign in.

> **⚠ Warning**
> After 5 consecutive failed login attempts, your account is temporarily locked with exponential backoff (up to 60 minutes). Contact a super_admin to unlock your account earlier.

5.3 Dashboard

The Dashboard is the first page shown after login. It provides a real-time overview of platform activity.

- Period selector: Switch between 7-day, 30-day, and 90-day views using the buttons in the top-right corner.
- KPI cards: New leads, Conversion rate, Average testimonial rating, Portfolio view count.
- Recent leads table: Shows the 5 most recent contact form submissions. Click any row to view the full lead.
- Recent activity feed: Shows the 10 most recent admin actions from the audit log.

- Media pending scan warning: A yellow banner appears if any uploaded media is awaiting virus scan.

5.4 Managing Leads

Lead status workflow

Status	Meaning	When to use
New	Recently submitted, unread	Default on submission
Read	Opened by an admin	Automatically set on first open
Replied	A response has been sent to the prospect	After sending a reply email
Converted	The lead became a paying client	After contract signed

To update a lead status: on the Leads page, find the lead in the table and click the status badge to open a dropdown. Select the new status. The update is applied immediately (optimistic update) and synced to the server.

Searching and filtering leads

- Use the search box (top-right of leads page) to search by name, email, or company name.
- Click a status filter pill (New, Read, Replied, Converted) to show only leads in that status.
- Both filters can be combined: e.g. filter by status = 'New' and search for 'tech'.

5.5 Managing Portfolio Projects

Creating a new project

17. Go to Admin → Portfolio → click New Project.
18. Fill in: Slug (URL-safe, e.g. brandhouse-rebrand), Title, Client Name, Category.
19. Optionally add: Description, Cover Image (from media library), Project URL, Completed Date.
20. Toggle Is Featured to highlight the project on the home page.
21. Toggle Is Published to make the project visible on the public portfolio page.
22. Click Save. The project slug determines its URL: /portfolio/{slug}.

> ⚠ **Warning**
> Slugs must be lowercase kebab-case (letters, numbers, and hyphens only). Example: brandhouse-agency-rebrand. A slug cannot be changed after publication without breaking existing links.

5.6 Managing Testimonials

The Testimonials page allows editors to create, update, delete, and reorder testimonials for the carousel.

- Drag and drop testimonial rows to change display order. Click Save Order to commit changes.
- The displayOrder field controls the sequence in the public carousel and the Testimonials page grid.
- Marking a testimonial as Featured ensures it appears on the home page carousel.

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- Video URLs must be from YouTube (youtube.com) or Vimeo (player.vimeo.com). Other URLs are rejected.

5.7 Media Library

Allowed type	Max size	After upload
JPEG, PNG, WebP, GIF	10 MB	Queued for virus scan
SVG	10 MB	Queued for virus scan
PDF	10 MB	Queued for virus scan
MP4, MOV	50 MB	Queued for virus scan

After upload, every file is scanned by ClamAV before its public URL is made available. Scan statuses:

- Scanning: The file has been uploaded but the scan has not completed.
- Clean: Scan passed. The file's public URL is available for use in portfolio and testimonial records.
- Infected: The file contained malware. It has been quarantined and cannot be published.
- Error: The scan process failed (e.g. Lambda timeout). The file will be retried. Contact DevOps if it stays in Error after 24 hours.

> **⚠ Warning**
> Before deleting a media asset, ensure it is not referenced by any portfolio project or testimonial. The API will reject the deletion request if references exist.

5.8 Enabling MFA on Your Account

23. Go to Admin → Settings.
24. Under Two-factor authentication, click Set up MFA.
25. Scan the QR code with Google Authenticator, Authy, or any TOTP-compatible app.
26. Alternatively, manually enter the secret key shown below the QR code.
27. Enter the 6-digit code from your authenticator app and click Verify & Enable.
28. MFA is now active. Future logins will require your password plus the current 6-digit code.

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6. Maintenance Plan

6.1 Routine Maintenance Schedule

Frequency	Task	Owner	Effort
Daily (automated)	Database backup (pg_dump → S3)	GitHub Actions / cron	0 min
Daily (automated)	Expired session cleanup (admin_sessions)	ECS cron job (02:00 UTC)	0 min
Daily	Review CloudWatch alarm emails	DevOps	10 min
Weekly	Review dependency updates (Dependabot PRs)	Engineering	30 min
Weekly	Audit log review for anomalous actions	DevOps / Admin	15 min
Monthly (automated)	Audit log purge (entries > 2 years)	ECS cron job (03:00 UTC, 1st)	0 min
Monthly	Review CloudWatch cost and usage report	DevOps	20 min
Monthly	Run pnpm audit --audit-level=critical on both repos	Engineering	30 min
Quarterly	Rotate JWT_SECRET and REVALIDATION_SECRET	DevOps	30 min
Quarterly	Review admin user access (remove leavers)	super_admin	15 min
Quarterly	Restore test: restore from backup to staging DB	DevOps	60 min
Annually	SSL certificate renewal review (ACM auto-renews)	DevOps	15 min
Annually	Review and update all admin passwords	All admins	15 min
Annually	Full disaster recovery drill	DevOps	4 hours

6.2 Dependency Management

Dependabot is configured to scan both the frontend and backend repositories weekly and open PRs for dependency updates. The policy is:

- Security patches (CRITICAL or HIGH CVE): Merge within 24 hours. The pnpm audit --audit-level=critical CI gate blocks all deploys until resolved.
- Minor and patch updates: Merge within 1 week. Review automated test results before merging.
- Major version updates: Evaluate manually. Create a dedicated branch and test thoroughly before merging.

6.3 Database Maintenance

Supabase Pro manages the following automatically:

- Daily automatic backups with 7-day point-in-time recovery.
- Vacuum and autovacuum operations.
- PostgreSQL minor version patching.

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Manual tasks required:

- Monitor table bloat in contact_leads and audit_log (run ANALYZE monthly).
- Verify new indexes are used after schema migrations (use EXPLAIN ANALYZE).
- Review pg_stat_activity for long-running queries quarterly.

6.4 Container Image Updates

ECR holds the last 10 tagged production images (policy applied by Terraform lifecycle rules). Untagged images are automatically expired after 7 days. To update the production image:

29. Merge changes to main branch.
30. GitHub Actions builds a new multi-arch image (amd64 + arm64) and runs Trivy scanning.
31. If Trivy passes, the image is pushed to ECR and tagged with the Git short SHA.
32. ECS performs a rolling deployment (minimum 50% healthy, maximum 200% capacity).
33. The circuit breaker monitors health checks. If two consecutive tasks fail their health check, ECS rolls back to the previous task definition automatically.

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7. Backup Strategy

7.1 What Is Backed Up

Data	Method	Frequency	Retention	Storage
PostgreSQL (Supabase)	Automatic (Supabase managed)	Continuous / Daily	7 days PITR	Supabase infrastructure
PostgreSQL (pg_dump)	Manual via backup-database.sh	Daily 02:00 UTC	365 days	S3 ap-south-1 + S3 us-east-1 (DR)
Media assets	Supabase Storage (managed)	Continuous	Until manually deleted	Supabase Storage
Sanity CMS content	Sanity managed + Git-versioned schemas	Continuous	Indefinite	Sanity CDN
Application code	Git (GitHub)	Per commit	Indefinite	GitHub
Infrastructure state	Terraform (S3 backend, versioned)	Per apply	Indefinite (versioned)	S3 ap-south-1
ALB access logs	S3 (auto by ALB)	Continuous	30 days	S3
API logs (CloudWatch)	CloudWatch Logs	Continuous	30 days	CloudWatch
Audit log (exported)	Weekly export to S3	Weekly	2 years (Glacier IR)	S3
Container images	ECR	Per push	10 tagged versions	ECR ap-south-1

7.2 Backup Process — Database

The automated daily backup is executed by the script at devops/scripts/backup-database.sh. The process is:

34. The script retrieves DATABASE_URL from AWS Secrets Manager (no plaintext credentials in scripts).
35. pg_dump is executed with --format=custom --compress=9 to produce a compressed binary dump optimised for pg_restore.
36. The dump file is uploaded to S3 with --sse aws:kms encryption before touching the wire.
37. The script verifies the uploaded object size is > 1 KB. If the size check fails, the script exits with a non-zero code and triggers an alert.
38. S3 cross-region replication automatically copies the object to the us-east-1 DR bucket within minutes.

7.3 S3 Lifecycle Tiers

Age	Storage class	Cost (relative)	Use case
0–7 days	STANDARD	$$$$	Immediate restore (same day)
7–30 days	STANDARD_IA	$$$	Quick restore (hours)
30–365 days	GLACIER_IR	$	Quarterly restore drill
> 365 days	Expires (deleted)	—	N/A

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7.4 Restore Procedure

Restoring the database from pg_dump backup

```
# List available backups
aws s3 ls s3://outpro-production-backups/db/ --recursive
```

```
# Download the desired backup
TIMESTAMP='20260413_020001'
aws s3 cp s3://outpro-production-backups/db/${TIMESTAMP}/database.dump
/tmp/restore.dump
```

```
# Restore (--clean drops objects before recreating)
pg_restore --clean --no-owner --no-acl -d "$DATABASE_URL" /tmp/restore.dump
```

✕ Important
Always test the restore on a staging database first. The --clean flag will drop and recreate all tables, overwriting any existing data in the target database.

7.5 Recovery Point and Time Objectives

Scenario	RPO (max data loss)	RTO (restore time)	Procedure
Data corruption (application bug)	Up to 24 hours	30–120 minutes	Restore from pg_dump or use Supabase PITR
Accidental data deletion	Up to 24 hours	30–60 minutes	Restore from previous pg_dump
ECS service failure	0 (stateless)	< 5 minutes	ECS auto-replaces failed tasks
AWS region outage (ap-south-1)	Up to 24 hours	2–4 hours	Restore from DR bucket in us-east-1
Vercel deployment failure	0 (atomic deploys)	< 1 minute	Vercel one-click rollback
Complete account compromise	Up to 24 hours	4–8 hours	Full DR playbook

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8. Troubleshooting

8.1 Frontend Issues

Symptom	Likely cause	Resolution
Contact form shows reCAPTCHA error	NEXT_PUBLIC_RECAPTCHA_SITE_KEY missing or wrong	Verify the env var in Vercel Dashboard → Settings → Environment Variables
Portfolio grid shows no projects	Backend API unreachable or CORS mismatch	Check NEXT_PUBLIC_API_URL; verify ALLOWED_ORIGINS in Secrets Manager includes the Vercel domain
Admin login redirects to /login immediately	JWT_SECRET mismatch between frontend and backend	Ensure the same JWT_SECRET is set in both Vercel env vars and AWS Secrets Manager
Images not loading (Supabase)	NEXT_PUBLIC_SUPABASE_URL missing or media not yet clean-scanned	Verify env var; check media asset scan status in admin panel
Stale content after CMS update	ISR not triggered or REVALIDATION_SECRET mismatch	Manually trigger revalidate: curl -X POST https://www.outpro.india/api/revalidate -H 'Authorization: Bearer YOUR_SECRET'
Page returns 500 in production	Build-time env var missing	Check Vercel deployment logs → Functions tab for error details

8.2 Backend / API Issues

Symptom	Likely cause	Resolution
All API requests return 503	ECS tasks not running	Check ECS service in AWS Console; run: aws ecs describe-services --cluster outpro-production-cluster --services outpro-production-api
API returns 401 on all requests	JWT_SECRET mismatch between ECS and frontend	Rotate secrets with devops/scripts/rotate-secrets.sh
POST /contact returns 403 CAPTCHA_FAILED	RECAPTCHA_SECRET_KEY wrong or score < 0.5	Verify the secret key in Secrets Manager; check Google reCAPTCHA admin for score distribution
Email notifications not delivered	RESEND_API_KEY expired or domain not verified	Check Resend dashboard; verify outpro.india domain DNS (SPF, DKIM) with: dig TXT outpro.india
CRM sync fails silently	HUBSPOT_API_KEY expired	Rotate the HubSpot private app token; update in Secrets Manager; force ECS redeployment
Database connection pool exhausted	Too many concurrent ECS tasks or slow queries	Check CloudWatch for ECS task count; review pg_stat_activity for long-running queries
Media upload returns 413	File exceeds size limit	Limit: 10 MB images, 50 MB videos. Client validates before upload; server validates on receive.

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Health check returns 503	Database unreachable	Verify Supabase status at status.supabase.com; check DATABASE_URL has not expired

8.3 Infrastructure Issues

Symptom	Likely cause	Resolution
ECS tasks keep restarting	Application crash on startup (bad env var)	View logs: aws logs tail /ecs/outpro-production-api --follow
ECS deployment stuck in IN_PROGRESS	New tasks failing health check	Check CloudWatch logs for startup errors; ECS circuit breaker will auto-rollback after threshold
ALB 5xx alarm firing	Application errors or OOM kills	Check CloudWatch ECS memory metric; increase api_memory in Terraform if hitting 100%
Terraform apply fails with state lock	Previous apply crashed	Run: terraform force-unlock LOCK_ID (find ID in the error message)
Cloudflare rate limit blocking legitimate traffic	Rate limit too aggressive	Adjust cloudflare_rate_limit threshold in Terraform; re-apply
ACM certificate not validating	DNS validation record missing	Re-run terraform apply to recreate the Cloudflare DNS validation record
S3 backup upload failing	ECS task role missing s3:PutObject permission	Verify aws_iam_role_policy.ecs_task_s3 is applied; check CloudTrail for access denied events

8.4 Viewing Logs

Live API log streaming

```
aws logs tail /ecs/outpro-production-api --follow
```

Filter for ERROR-level messages only

```
aws logs tail /ecs/outpro-production-api --follow --filter-pattern "[timestamp,
level=ERROR, ...]"
```

Search for a specific request ID

```
aws logs filter-log-events --log-group-name /ecs/outpro-production-api --filter-
pattern "{$.requestId = \"YOUR-REQUEST-ID\"}"
```

8.5 Emergency Procedures

Emergency	Immediate action	Command / location
API completely down	Roll back to previous ECS task definition	bash devops/scripts/rollback.sh production
Suspected credential compromise	Rotate all secrets immediately	bash devops/scripts/rotate-secrets.sh production

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DDoS attack	Enable Cloudflare Under Attack mode	Cloudflare Dashboard → Security → Settings → I'm Under Attack
Data breach suspected	Take ECS service to 0 tasks (isolate)	aws ecs update-service --cluster ... --desired-count 0
Accidental data deletion	Restore from latest pg_dump backup	See Section 7.4 Restore Procedure
Production deploy broken	Rollback Vercel deployment	Vercel Dashboard → Deployments → Previous deploy → Promote to Production

8.6 Escalation Path

Severity	Definition	Response time	Contact
P1 – Critical	Site is down or data breach	< 15 minutes	devops@outpro.india + phone
P2 – High	Core feature broken (contact form, admin login)	< 1 hour	devops@outpro.india
P3 – Medium	Non-critical feature degraded	< 4 hours	GitHub issue + Slack
P4 – Low	Minor bug or cosmetic issue	Next sprint	GitHub issue

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Appendix A — API Endpoint Reference

Base URL: https://api.outpro.india/api/v1 — All responses use the standard envelope: { success, data, meta } or { success, error }.

#	Method	Endpoint	Auth	Description
1	POST	/auth/login	Public	Admin login + TOTP verification
2	POST	/auth/logout	JWT	Logout and revoke session
3	GET	/auth/me	JWT	Get authenticated user profile
4	PATCH	/auth/password	JWT	Change password
5	POST	/auth/mfa/setup	JWT	Generate MFA QR code
6	POST	/auth/mfa/verify	JWT	Enable MFA on account
7	POST	/contact	Public + reCAPTCHA	Submit contact form
8	GET	/contact/leads	JWT	List leads (paginated)
9	GET	/contact/leads/stats	JWT	Lead statistics
10	GET	/contact/leads/:id	JWT	Get single lead
11	PATCH	/contact/leads/:id	JWT editor+	Update lead status
12	DELETE	/contact/leads/:id	JWT super_admin	Delete lead
13	GET	/portfolio	Public	List portfolio projects
14	GET	/portfolio/categories	Public	Category project counts
15	GET	/portfolio/:slug	Public	Get project by slug
16	POST	/portfolio	JWT editor+	Create project
17	PATCH	/portfolio/:id	JWT editor+	Update project
18	DELETE	/portfolio/:id	JWT super_admin	Delete project
19	POST	/portfolio/revalidate	Webhook secret	Trigger Next.js ISR
20	GET	/testimonials	Public	List testimonials
21	GET	/testimonials/stats	JWT	Testimonial statistics
22	GET	/testimonials/:id	Public	Get testimonial
23	POST	/testimonials	JWT editor+	Create testimonial
24	PATCH	/testimonials/reorder	JWT editor+	Batch reorder carousel
25	PATCH	/testimonials/:id	JWT editor+	Update testimonial
26	DELETE	/testimonials/:id	JWT super_admin	Delete testimonial
27	GET	/admin/dashboard	JWT	KPI dashboard (7d/30d/90d)
28	GET	/admin/users	JWT super_admin	List admin users
29	POST	/admin/users	JWT super_admin	Create admin user

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30	PATCH	/admin/users/:id	JWT super_admin	Update admin user
31	DELETE	/admin/users/:id	JWT super_admin	Delete admin user
32	GET	/admin/audit-log	JWT	View audit trail
33	POST	/admin/media	JWT editor+	Upload media asset
34	GET	/admin/media	JWT	List media assets
35	DELETE	/admin/media/:id	JWT super_admin	Delete media asset
36	GET	/health	Public	Uptime health probe

Appendix B — Environment Variable Reference

Variable	Service	Required	Description
DATABASE_URL	Backend	✓	PostgreSQL connection URL (Supabase)
JWT_SECRET	Backend + Frontend	✓	Min 64 chars — generate: openssl rand -hex 64
JWT_EXPIRY	Backend	✓	Token lifetime (default: 8h)
REVALIDATION_SECRET	Backend + Frontend	✓	Min 32 chars — Sanity ISR webhook
RECAPTCHA_SECRET_KEY	Backend	✓	Server-side reCAPTCHA v3 secret
NEXT_PUBLIC_RECAPTCHA_SITE_KEY	Frontend	✓	Client-side reCAPTCHA v3 key
RESEND_API_KEY	Backend	✓	Transactional email API key
EMAIL_FROM	Backend	✓	Sender address (default: noreply@outpro.india)
EMAIL_ADMIN_NOTIFY	Backend	✓	Lead notification recipient
SUPABASE_URL	Backend + Frontend	✓	Supabase project URL
SUPABASE_SERVICE_ROLE_KEY	Backend	✓	Bypasses RLS — keep secret
NEXT_PUBLIC_SUPABASE_URL	Frontend	✓	Public Supabase URL for image URLs
NEXT_PUBLIC_API_URL	Frontend	✓	Backend API base URL
NEXT_PUBLIC_SANITY_PROJECT_ID	Frontend	✓	Sanity project ID
NEXT_PUBLIC_SANITY_DATASET	Frontend	✓	Sanity dataset (default: production)
SANITY_API_TOKEN	Frontend	✓	Sanity read token (server-side only)
HUBSPOT_API_KEY	Backend	✗	CRM sync — skipped if absent
UPSTASH_REDIS_REST_URL	Backend	✗	Rate limiting — falls back to memory
UPSTASH_REDIS_REST_TOKEN	Backend	✗	Upstash Redis auth token

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BCRYPT_ROUNDS	Backend	✗	Password hash cost (default: 12)
LOG_LEVEL	Backend	✗	error/warn/info/debug (default: info)
NEXT_PUBLIC_GA4_MEASUREMENT_ID	Frontend	✗	Google Analytics 4 tracking ID

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